

CS-GY 6763: Lecture 4

High-Dimensional Geometry

NYU, Prof. Ainesh Bakshi

UNIFYING THEME OF THE COURSE

How do we work with data represented as high-dimensional vectors?

- High-dimensional similarity search.

How do we work with data represented as high-dimensional vectors?

- High-dimensional similarity search.
- Iterative methods for optimizing functions in high dimensions.

UNIFYING THEME OF THE COURSE

How do we work with data represented as high-dimensional vectors?

- High-dimensional similarity search.
- Iterative methods for optimizing functions in high dimensions.
- SVD + low-rank approximation to find and visualize low-dimensional structure.

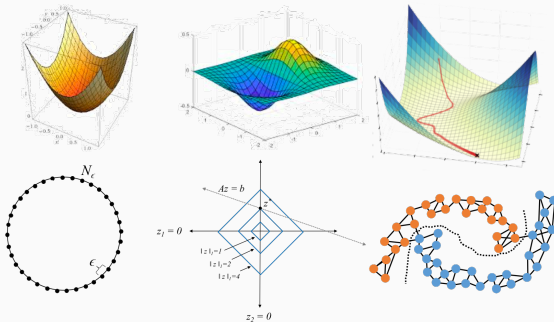
UNIFYING THEME OF THE COURSE

How do we work with data represented as high-dimensional vectors?

- High-dimensional similarity search.
- Iterative methods for optimizing functions in high dimensions.
- SVD + low-rank approximation to find and visualize low-dimensional structure.
- Represent large graphs as high-dimensional vectors to uncover structure.

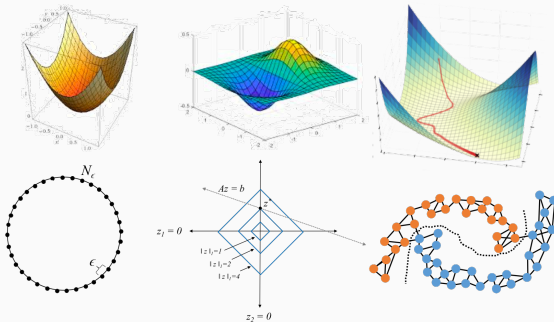
HIGH-DIMENSIONAL SPACE IS DIFFERENT

We often visualize data and algorithms in one, two, or three dimensions.



HIGH-DIMENSIONAL SPACE IS DIFFERENT

We often visualize data and algorithms in one, two, or three dimensions.



This lecture: High-dimensional space can look **very different** from low-dimensional space, so these pictures can be misleading.

ORTHOGONAL VECTORS

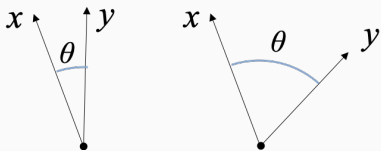
Recall the inner product between two d -dimensional vectors:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x} = \sum_{j=1}^d x[j]y[j]$$

ORTHOGONAL VECTORS

Recall the inner product between two d -dimensional vectors:

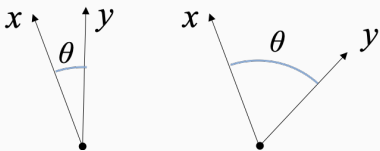
$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x} = \sum_{j=1}^d x[j]y[j]$$



ORTHOGONAL VECTORS

Recall the inner product between two d -dimensional vectors:

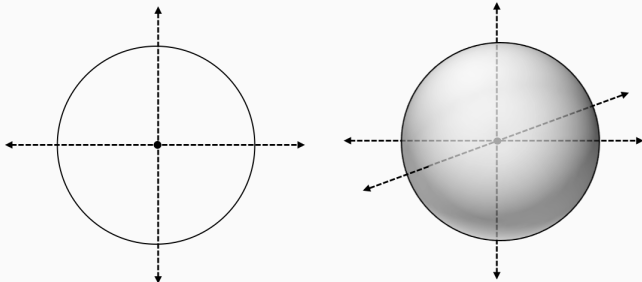
$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x} = \sum_{j=1}^d x[j]y[j]$$



$$\langle \mathbf{x}, \mathbf{y} \rangle = \cos(\theta) \cdot \|\mathbf{x}\|_2 \cdot \|\mathbf{y}\|_2$$

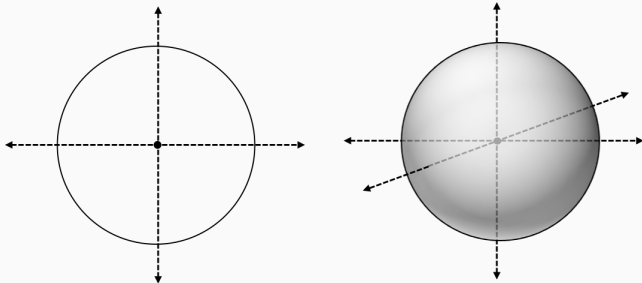
ORTHOGONAL VECTORS

What is the largest possible number of **mutually orthogonal** unit vectors $\mathbf{x}_1, \dots, \mathbf{x}_t$ in d -dimensional space? That is, $|\mathbf{x}_i^T \mathbf{x}_j| = 0$ for all distinct i, j .



ORTHOGONAL VECTORS

What is the largest possible number of **nearly orthogonal** unit vectors $\mathbf{x}_1, \dots, \mathbf{x}_t$ in two or three dimensions? That is, $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for all distinct i, j . Consider a constant such as $\epsilon = 1/10$.



ORTHOGONAL VECTORS

What is the largest possible number of **nearly orthogonal** unit vectors $\mathbf{x}_1, \dots, \mathbf{x}_t$ in d dimensions? That is, $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for all distinct i, j , where ϵ is a constant such as $1/10$.

1. d 2. d^2 3. $2^{\sqrt{d}}$ 4. $2^{\Theta(d)}$

ORTHOGONAL VECTORS

Formal claim: There are absolute constants $c, C > 0$ such that, for every $0 < \epsilon < 1$ and $d \geq C/\epsilon^2$, there is a set of at least $2^{c\epsilon^2 d}$ unit vectors in $\mathbb{R}^d$ such that $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for all distinct i, j .

ORTHOGONAL VECTORS

Formal claim: There are absolute constants $c, C > 0$ such that, for every $0 < \epsilon < 1$ and $d \geq C/\epsilon^2$, there is a set of at least $2^{c\epsilon^2 d}$ unit vectors in $\mathbb{R}^d$ such that $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for all distinct i, j .

Proof strategy: Use the **probabilistic method**. Generate $t = 2^{\Omega(\epsilon^2 d)}$ random unit vectors that are unlikely to have large absolute inner products.

ORTHOGONAL VECTORS

Formal claim: There are absolute constants $c, C > 0$ such that, for every $0 < \epsilon < 1$ and $d \geq C/\epsilon^2$, there is a set of at least $2^{c\epsilon^2 d}$ unit vectors in $\mathbb{R}^d$ such that $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for all distinct i, j .

Proof strategy: Use the **probabilistic method**. Generate $t = 2^{\Omega(\epsilon^2 d)}$ random unit vectors that are unlikely to have large absolute inner products.

1. Show that $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for all distinct i, j with positive probability.

ORTHOGONAL VECTORS

Formal claim: There are absolute constants $c, C > 0$ such that, for every $0 < \epsilon < 1$ and $d \geq C/\epsilon^2$, there is a set of at least $2^{c\epsilon^2 d}$ unit vectors in $\mathbb{R}^d$ such that $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for all distinct i, j .

Proof strategy: Use the **probabilistic method**. Generate $t = 2^{\Omega(\epsilon^2 d)}$ random unit vectors that are unlikely to have large absolute inner products.

1. Show that $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for all distinct i, j with positive probability.
2. Conclude that some set of t unit vectors has all pairwise absolute inner products bounded by ϵ .

PROBABILISTIC METHOD

Claim: When $\epsilon^2 d$ is sufficiently large, there are at least $2^{\Omega(\epsilon^2 d)}$ nearly orthogonal unit vectors in d -dimensional space.

PROBABILISTIC METHOD

Claim: When $\epsilon^2 d$ is sufficiently large, there are at least $2^{\Omega(\epsilon^2 d)}$ nearly orthogonal unit vectors in d -dimensional space.

Proof: Let all td entries of $\mathbf{x}_1, \dots, \mathbf{x}_t$ be mutually independent, with each entry equal to $\pm 1/\sqrt{d}$ with probability $1/2$. For example,

$$\mathbf{x}_1 = \left(\frac{1}{\sqrt{d}}, \frac{1}{\sqrt{d}}, -\frac{1}{\sqrt{d}}, \frac{1}{\sqrt{d}}, \dots, \frac{1}{\sqrt{d}} \right).$$

PROBABILISTIC METHOD

For fixed distinct i, j : Let $\mathbf{x}_1, \dots, \mathbf{x}_t$ have mutually independent entries, each equal to $\pm 1/\sqrt{d}$ with probability $1/2$.

- $\|\mathbf{x}_i\|_2^2 =$

PROBABILISTIC METHOD

For fixed distinct i, j : Let $\mathbf{x}_1, \dots, \mathbf{x}_t$ have mutually independent entries, each equal to $\pm 1/\sqrt{d}$ with probability $1/2$.

- $\|\mathbf{x}_i\|_2^2 = \sum_{k=1}^d \frac{1}{d} = 1$

PROBABILISTIC METHOD

For fixed distinct i, j : Let $\mathbf{x}_1, \dots, \mathbf{x}_t$ have mutually independent entries, each equal to $\pm 1/\sqrt{d}$ with probability $1/2$.

- $\|\mathbf{x}_i\|_2^2 = \sum_{k=1}^d \frac{1}{d} = 1$

- $\mathbb{E}[\mathbf{x}_i^T \mathbf{x}_j] =$

PROBABILISTIC METHOD

For fixed distinct i, j : Let $\mathbf{x}_1, \dots, \mathbf{x}_t$ have mutually independent entries, each equal to $\pm 1/\sqrt{d}$ with probability $1/2$.

- $\|\mathbf{x}_i\|_2^2 = \sum_{k=1}^d \frac{1}{d} = 1$
- $\mathbb{E}[\mathbf{x}_i^T \mathbf{x}_j] = \sum_{k=1}^d \mathbb{E}[\mathbf{x}_i[k] \mathbf{x}_j[k]] = \sum_{k=1}^d \mathbb{E}[\mathbf{x}_i[k]] \mathbb{E}[\mathbf{x}_j[k]] = 0$

PROBABILISTIC METHOD

For fixed distinct i, j : Let $\mathbf{x}_1, \dots, \mathbf{x}_t$ have mutually independent entries, each equal to $\pm 1/\sqrt{d}$ with probability $1/2$.

- $\|\mathbf{x}_i\|_2^2 = \sum_{k=1}^d \frac{1}{d} = 1$
- $\mathbb{E}[\mathbf{x}_i^T \mathbf{x}_j] = \sum_{k=1}^d \mathbb{E}[\mathbf{x}_i[k] \mathbf{x}_j[k]] = \sum_{k=1}^d \mathbb{E}[\mathbf{x}_i[k]] \mathbb{E}[\mathbf{x}_j[k]] = 0$
- $\mathbf{x}_i[k] \mathbf{x}_j[k] = \begin{cases} +1/d & \text{w.p. } 1/2 \\ -1/d & \text{w.p. } 1/2 \end{cases}$

PROBABILISTIC METHOD

For fixed distinct i, j : Let $\mathbf{x}_1, \dots, \mathbf{x}_t$ have mutually independent entries, each equal to $\pm 1/\sqrt{d}$ with probability $1/2$.

- $\|\mathbf{x}_i\|_2^2 = \sum_{k=1}^d \frac{1}{d} = 1$
- $\mathbb{E}[\mathbf{x}_i^T \mathbf{x}_j] = \sum_{k=1}^d \mathbb{E}[\mathbf{x}_i[k] \mathbf{x}_j[k]] = \sum_{k=1}^d \mathbb{E}[\mathbf{x}_i[k]] \mathbb{E}[\mathbf{x}_j[k]] = 0$
- $\mathbf{x}_i[k] \mathbf{x}_j[k] = \begin{cases} +1/d & \text{w.p. } 1/2 \\ -1/d & \text{w.p. } 1/2 \end{cases}$
- $\text{Var}[\mathbf{x}_i^T \mathbf{x}_j] =$

PROBABILISTIC METHOD

For fixed distinct i, j : Let $\mathbf{x}_1, \dots, \mathbf{x}_t$ have mutually independent entries, each equal to $\pm 1/\sqrt{d}$ with probability $1/2$.

- $\|\mathbf{x}_i\|_2^2 = \sum_{k=1}^d \frac{1}{d} = 1$
- $\mathbb{E}[\mathbf{x}_i^T \mathbf{x}_j] = \sum_{k=1}^d \mathbb{E}[\mathbf{x}_i[k] \mathbf{x}_j[k]] = \sum_{k=1}^d \mathbb{E}[\mathbf{x}_i[k]] \mathbb{E}[\mathbf{x}_j[k]] = 0$
- $\mathbf{x}_i[k] \mathbf{x}_j[k] = \begin{cases} +1/d & \text{w.p. } 1/2 \\ -1/d & \text{w.p. } 1/2 \end{cases}$
- $\text{Var}[\mathbf{x}_i^T \mathbf{x}_j] = \sum_{k=1}^d \text{Var}[\mathbf{x}_i[k] \mathbf{x}_j[k]] = \sum_{k=1}^d \frac{1}{d^2} = \frac{1}{d}$

INFORMAL PROOF

For fixed distinct i, j , let $Z = \mathbf{x}_i^T \mathbf{x}_j = \sum_{k=1}^d C_k$, where the C_k are independent and uniform on $\{-1/d, +1/d\}$.

INFORMAL PROOF

For fixed distinct i, j , let $Z = \mathbf{x}_i^T \mathbf{x}_j = \sum_{k=1}^d C_k$, where the C_k are independent and uniform on $\{-1/d, +1/d\}$.

The central limit theorem suggests Gaussian behavior near the mean. For the tail we need, Hoeffding's inequality gives the rigorous bound

$$\Pr[|Z| \geq \epsilon] \leq 2e^{-\epsilon^2 d/2}.$$

INFORMAL PROOF

For fixed distinct i, j , let $Z = \mathbf{x}_i^T \mathbf{x}_j = \sum_{k=1}^d C_k$, where the C_k are independent and uniform on $\{-1/d, +1/d\}$.

The central limit theorem suggests Gaussian behavior near the mean. For the tail we need, Hoeffding's inequality gives the rigorous bound

$$\Pr[|Z| \geq \epsilon] \leq 2e^{-\epsilon^2 d/2}.$$

A union bound over all distinct pairs gives

INFORMAL PROOF

For fixed distinct i, j , let $Z = \mathbf{x}_i^T \mathbf{x}_j = \sum_{k=1}^d C_k$, where the C_k are independent and uniform on $\{-1/d, +1/d\}$.

The central limit theorem suggests Gaussian behavior near the mean. For the tail we need, Hoeffding's inequality gives the rigorous bound

$$\Pr[|Z| \geq \epsilon] \leq 2e^{-\epsilon^2 d/2}.$$

A union bound over all distinct pairs gives

$$\Pr[\exists i < j : |\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2 \binom{t}{2} e^{-\epsilon^2 d/2}.$$

INFORMAL PROOF

For fixed distinct i, j , let $Z = \mathbf{x}_i^T \mathbf{x}_j = \sum_{k=1}^d C_k$, where the C_k are independent and uniform on $\{-1/d, +1/d\}$.

The central limit theorem suggests Gaussian behavior near the mean. For the tail we need, Hoeffding's inequality gives the rigorous bound

$$\Pr[|Z| \geq \epsilon] \leq 2e^{-\epsilon^2 d/2}.$$

A union bound over all distinct pairs gives

$$\Pr[\exists i < j : |\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2 \binom{t}{2} e^{-\epsilon^2 d/2}.$$

When $\epsilon^2 d$ is sufficiently large, taking $t = \left\lfloor \frac{1}{4} e^{\epsilon^2 d/4} \right\rfloor$ makes the failure probability at most $1/16$. Thus $t = e^{\Omega(\epsilon^2 d)}$ vectors exist.

Use an exponential concentration inequality!

Theorem (Chernoff Bound)

Let $X_1, X_2, \dots, X_d$ be independent $\{0, 1\}$ -valued random variables and let $S = \sum_{i=1}^d X_i$. For any $0 < \epsilon \leq 1$,

$$\Pr[|S - \mathbb{E}[S]| \geq \epsilon \mathbb{E}[S]] \leq 2e^{\frac{-\epsilon^2 \mathbb{E}[S]}{3}}.$$

FORMAL PROOF

Use an exponential concentration inequality!

Theorem (Chernoff Bound)

Let $X_1, X_2, \dots, X_d$ be independent $\{0, 1\}$ -valued random variables and let $S = \sum_{i=1}^d X_i$. For any $0 < \epsilon \leq 1$,

$$\Pr[|S - \mathbb{E}[S]| \geq \epsilon \mathbb{E}[S]] \leq 2e^{\frac{-\epsilon^2 \mathbb{E}[S]}{3}}.$$

This does not immediately apply because our random variables are $\pm 1/d$, rather than 0 or 1. A common trick is to shift and scale them.

FORMAL PROOF

For each k , define $B_k = (dC_k + 1)/2$. Then $B_1, \dots, B_d$ are independent Bernoulli($1/2$) random variables and $C_k = (2B_k - 1)/d$.

$$Z = \sum_{k=1}^d C_k = \frac{1}{d} \sum_{k=1}^d (2B_k - 1)$$

FORMAL PROOF

For each k , define $B_k = (dC_k + 1)/2$. Then $B_1, \dots, B_d$ are independent Bernoulli($1/2$) random variables and $C_k = (2B_k - 1)/d$.

$$\begin{aligned} Z &= \sum_{k=1}^d C_k = \frac{1}{d} \sum_{k=1}^d (2B_k - 1) \\ &= \frac{2}{d} \left(\sum_{k=1}^d B_k - \frac{d}{2} \right). \end{aligned}$$

FORMAL PROOF

For each k , define $B_k = (dC_k + 1)/2$. Then $B_1, \dots, B_d$ are independent Bernoulli($1/2$) random variables and $C_k = (2B_k - 1)/d$.

$$\begin{aligned} Z &= \sum_{k=1}^d C_k = \frac{1}{d} \sum_{k=1}^d (2B_k - 1) \\ &= \frac{2}{d} \left(\sum_{k=1}^d B_k - \frac{d}{2} \right). \end{aligned}$$

This converts the problem to concentration of a sum of independent binary random variables.

FORMAL PROOF

Let $S = \sum_{k=1}^d B_k$, so $\mathbb{E}[S] = d/2$. Then

$$Z = \frac{2}{d} \left(S - \frac{d}{2} \right)$$

FORMAL PROOF

Let $S = \sum_{k=1}^d B_k$, so $\mathbb{E}[S] = d/2$. Then

$$Z = \frac{2}{d} \left(S - \frac{d}{2} \right)$$

$$\Pr[|Z| \geq \epsilon] = \Pr \left[\left| S - \frac{d}{2} \right| \geq \frac{\epsilon d}{2} \right]$$

FORMAL PROOF

Let $S = \sum_{k=1}^d B_k$, so $\mathbb{E}[S] = d/2$. Then

$$Z = \frac{2}{d} \left(S - \frac{d}{2} \right)$$

$$\begin{aligned} \Pr[|Z| \geq \epsilon] &= \Pr \left[\left| S - \frac{d}{2} \right| \geq \frac{\epsilon d}{2} \right] \\ &= \Pr[|S - \mathbb{E}[S]| \geq \epsilon \mathbb{E}[S]]. \end{aligned}$$

CHERNOFF BOUND

Theorem (Chernoff Bound)

Let $X_1, X_2, \dots, X_d$ be independent $\{0, 1\}$ -valued random variables and let $S = \sum_{i=1}^d X_i$. For any $0 < \epsilon \leq 1$,

$$\Pr[|S - \mathbb{E}[S]| \geq \epsilon \mathbb{E}[S]] \leq 2e^{\frac{-\epsilon^2 \mathbb{E}[S]}{3}}.$$

CHERNOFF BOUND

Theorem (Chernoff Bound)

Let $X_1, X_2, \dots, X_d$ be independent $\{0, 1\}$ -valued random variables and let $S = \sum_{i=1}^d X_i$. For any $0 < \epsilon \leq 1$,

$$\Pr[|S - \mathbb{E}[S]| \geq \epsilon \mathbb{E}[S]] \leq 2e^{\frac{-\epsilon^2 \mathbb{E}[S]}{3}}.$$

Apply the theorem to $B_1, \dots, B_d$, for which $\mathbb{E}[S] = d/2$:

CHERNOFF BOUND

Theorem (Chernoff Bound)

Let $X_1, X_2, \dots, X_d$ be independent $\{0, 1\}$ -valued random variables and let $S = \sum_{i=1}^d X_i$. For any $0 < \epsilon \leq 1$,

$$\Pr[|S - \mathbb{E}[S]| \geq \epsilon \mathbb{E}[S]] \leq 2e^{\frac{-\epsilon^2 \mathbb{E}[S]}{3}}.$$

Apply the theorem to $B_1, \dots, B_d$, for which $\mathbb{E}[S] = d/2$:

$$\Pr[|S - \mathbb{E}[S]| \geq \epsilon \mathbb{E}[S]] \leq 2e^{-\epsilon^2 \mathbb{E}[S]/3}$$

CHERNOFF BOUND

Theorem (Chernoff Bound)

Let $X_1, X_2, \dots, X_d$ be independent $\{0, 1\}$ -valued random variables and let $S = \sum_{i=1}^d X_i$. For any $0 < \epsilon \leq 1$,

$$\Pr[|S - \mathbb{E}[S]| \geq \epsilon \mathbb{E}[S]] \leq 2e^{\frac{-\epsilon^2 \mathbb{E}[S]}{3}}.$$

Apply the theorem to $B_1, \dots, B_d$, for which $\mathbb{E}[S] = d/2$:

$$\begin{aligned}\Pr[|S - \mathbb{E}[S]| \geq \epsilon \mathbb{E}[S]] &\leq 2e^{-\epsilon^2 \mathbb{E}[S]/3} \\ &= 2e^{-\epsilon^2 d/6}.\end{aligned}$$

Conclusion from Chernoff bound: For every fixed pair $i < j$,

$$\Pr[|\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2e^{-\epsilon^2 d/6}.$$

PROBABILISTIC METHOD

Conclusion from Chernoff bound: For every fixed pair $i < j$,

$$\Pr[|\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2e^{-\epsilon^2 d/6}.$$

By a union bound:

Conclusion from Chernoff bound: For every fixed pair $i < j$,

$$\Pr[|\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2e^{-\epsilon^2 d/6}.$$

By a union bound:

$$\Pr[\exists i < j : |\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2 \binom{t}{2} e^{-\epsilon^2 d/6}.$$

PROBABILISTIC METHOD

Conclusion from Chernoff bound: For every fixed pair $i < j$,

$$\Pr[|\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2e^{-\epsilon^2 d/6}.$$

By a union bound:

$$\Pr[\exists i < j : |\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2 \binom{t}{2} e^{-\epsilon^2 d/6}.$$

When $\epsilon^2 d$ is sufficiently large, set $t = \left\lfloor \frac{1}{4} e^{\epsilon^2 d/12} \right\rfloor$. Then the failure probability is at most $1/16$, so all pairs satisfy the desired bound with probability greater than 0.9.

PROBABILISTIC METHOD

Conclusion from Chernoff bound: For every fixed pair $i < j$,

$$\Pr[|\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2e^{-\epsilon^2 d/6}.$$

By a union bound:

$$\Pr[\exists i < j : |\mathbf{x}_i^T \mathbf{x}_j| \geq \epsilon] \leq 2 \binom{t}{2} e^{-\epsilon^2 d/6}.$$

When $\epsilon^2 d$ is sufficiently large, set $t = \left\lfloor \frac{1}{4} e^{\epsilon^2 d/12} \right\rfloor$. Then the failure probability is at most $1/16$, so all pairs satisfy the desired bound with probability greater than 0.9.

Therefore, a set of $t = e^{\Omega(\epsilon^2 d)} = 2^{\Omega(\epsilon^2 d)}$ such unit vectors exists.

ORTHOGONAL VECTORS

Final result: When $\epsilon^2 d$ is sufficiently large, there are at least $2^{\Omega(\epsilon^2 d)}$ unit vectors in d -dimensional space with $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for every distinct pair.

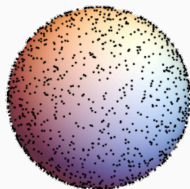
Corollary of proof: Independent random sign vectors sampled as above tend to be far apart and roughly equidistant in high dimensions.

ORTHOGONAL VECTORS

Final result: When $\epsilon^2 d$ is sufficiently large, there are at least $2^{\Omega(\epsilon^2 d)}$ unit vectors in d -dimensional space with $|\mathbf{x}_i^T \mathbf{x}_j| \leq \epsilon$ for every distinct pair.

Corollary of proof: Independent random sign vectors sampled as above tend to be far apart and roughly equidistant in high dimensions.

$$\|\mathbf{x}_i - \mathbf{x}_j\|_2^2 = 2 - 2\langle \mathbf{x}_i, \mathbf{x}_j \rangle \in [2 - 2\epsilon, 2 + 2\epsilon].$$

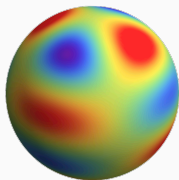


CURSE OF DIMENSIONALITY

Curse of dimensionality: Let $\mathbf{x}$ be uniform on $\mathcal{B}_d$ and place a query at the origin. For $0 < r < 1$,

$$\Pr[\|\mathbf{x}\|_2 \leq r] = r^d.$$

Thus $\Theta(r^{-d})$ independent samples are needed for a constant probability of finding one within distance r .

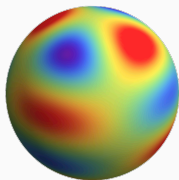


CURSE OF DIMENSIONALITY

Curse of dimensionality: Let $\mathbf{x}$ be uniform on $\mathcal{B}_d$ and place a query at the origin. For $0 < r < 1$,

$$\Pr[\|\mathbf{x}\|_2 \leq r] = r^d.$$

Thus $\Theta(r^{-d})$ independent samples are needed for a constant probability of finding one within distance r .



Low-dimensional or other structural assumptions can avoid this dependence.

Thinking in High Dimensions

UNIT BALL IN HIGH DIMENSIONS

Let $\mathcal{B}_d$ be the unit ball in d dimensions:

$$\mathcal{B}_d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}.$$

UNIT BALL IN HIGH DIMENSIONS

Let $\mathcal{B}_d$ be the unit ball in d dimensions:

$$\mathcal{B}_d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}.$$

A ball of radius R has volume $\frac{\pi^{d/2}}{\Gamma(d/2+1)} R^d$.

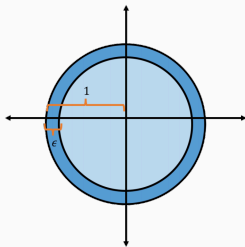
UNIT BALL IN HIGH DIMENSIONS

Let $\mathcal{B}_d$ be the unit ball in d dimensions:

$$\mathcal{B}_d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}.$$

A ball of radius R has volume $\frac{\pi^{d/2}}{\Gamma(d/2+1)} R^d$.

What fraction of the volume of $\mathcal{B}_d$ lies within ϵ of its surface?



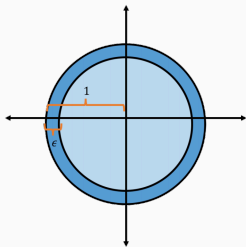
UNIT BALL IN HIGH DIMENSIONS

Let $\mathcal{B}_d$ be the unit ball in d dimensions:

$$\mathcal{B}_d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}.$$

A ball of radius R has volume $\frac{\pi^{d/2}}{\Gamma(d/2+1)} R^d$.

What fraction of the volume of $\mathcal{B}_d$ lies within ϵ of its surface?



$$\frac{\text{vol}(\mathcal{B}_d) - \text{vol}((1 - \epsilon)\mathcal{B}_d)}{\text{vol}(\mathcal{B}_d)} = 1 - (1 - \epsilon)^d$$

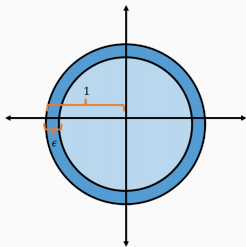
UNIT BALL IN HIGH DIMENSIONS

Let $\mathcal{B}_d$ be the unit ball in d dimensions:

$$\mathcal{B}_d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}.$$

A ball of radius R has volume $\frac{\pi^{d/2}}{\Gamma(d/2+1)} R^d$.

What fraction of the volume of $\mathcal{B}_d$ lies within ϵ of its surface?



$$\begin{aligned} \frac{\text{vol}(\mathcal{B}_d) - \text{vol}((1 - \epsilon)\mathcal{B}_d)}{\text{vol}(\mathcal{B}_d)} &= 1 - (1 - \epsilon)^d \\ &= 1 - ((1 - \epsilon)^{1/\epsilon})^{\epsilon d} \end{aligned}$$

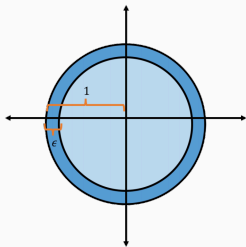
UNIT BALL IN HIGH DIMENSIONS

Let $\mathcal{B}_d$ be the unit ball in d dimensions:

$$\mathcal{B}_d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}.$$

A ball of radius R has volume $\frac{\pi^{d/2}}{\Gamma(d/2+1)} R^d$.

What fraction of the volume of $\mathcal{B}_d$ lies within ϵ of its surface?



$$\begin{aligned} \frac{\text{vol}(\mathcal{B}_d) - \text{vol}((1 - \epsilon)\mathcal{B}_d)}{\text{vol}(\mathcal{B}_d)} &= 1 - (1 - \epsilon)^d \\ &= 1 - ((1 - \epsilon)^{1/\epsilon})^{\epsilon d} \\ &\geq 1 - e^{-\epsilon d}, \quad 0 < \epsilon < 1. \end{aligned}$$

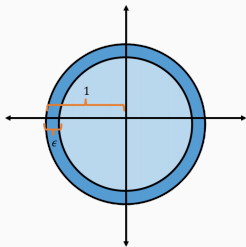
UNIT BALL IN HIGH DIMENSIONS

Let $\mathcal{B}_d$ be the unit ball in d dimensions:

$$\mathcal{B}_d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 \leq 1\}.$$

A ball of radius R has volume $\frac{\pi^{d/2}}{\Gamma(d/2+1)} R^d$.

What fraction of the volume of $\mathcal{B}_d$ lies within ϵ of its surface?



$$\begin{aligned} \frac{\text{vol}(\mathcal{B}_d) - \text{vol}((1 - \epsilon)\mathcal{B}_d)}{\text{vol}(\mathcal{B}_d)} &= 1 - (1 - \epsilon)^d \\ &= 1 - ((1 - \epsilon)^{1/\epsilon})^{\epsilon d} \\ &\geq 1 - e^{-\epsilon d}, \quad 0 < \epsilon < 1. \end{aligned}$$

Most of the volume of the unit ball lies near its surface!

ISOPERIMETRIC INEQUALITY

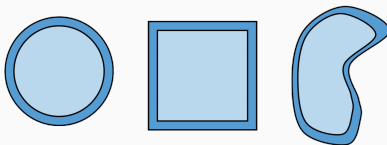
All but at most an $e^{-\epsilon d}$ fraction of the unit ball's volume lies within ϵ of its surface.

Isoperimetric inequality: Among sets with a fixed volume, the Euclidean ball minimizes surface area.

ISOPERIMETRIC INEQUALITY

All but at most an $e^{-\epsilon d}$ fraction of the unit ball's volume lies within ϵ of its surface.

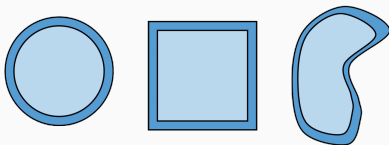
Isoperimetric inequality: Among sets with a fixed volume, the Euclidean ball minimizes surface area.



ISOPERIMETRIC INEQUALITY

All but at most an $e^{-\epsilon^d}$ fraction of the unit ball's volume lies within ϵ of its surface.

Isoperimetric inequality: Among sets with a fixed volume, the Euclidean ball minimizes surface area.

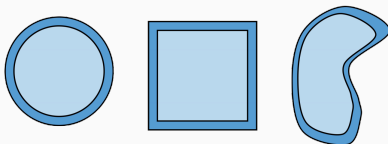


- A uniformly sampled point from a high-dimensional unit ball is very likely to lie near the boundary.

ISOPERIMETRIC INEQUALITY

All but at most an $e^{-\epsilon d}$ fraction of the unit ball's volume lies within ϵ of its surface.

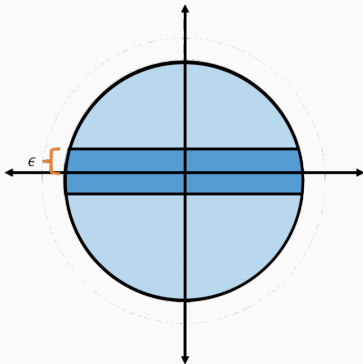
Isoperimetric inequality: Among sets with a fixed volume, the Euclidean ball minimizes surface area.



- A uniformly sampled point from a high-dimensional unit ball is very likely to lie near the boundary.
- Similar concentration holds for many regular high-dimensional bodies, but extra assumptions are necessary.

SLICES OF THE UNIT BALL

What fraction of the volume of $\mathcal{B}_d$ lies within ϵ of its equator?



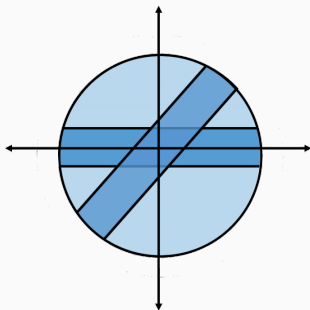
$$S = \{\mathbf{x} \in \mathcal{B}_d : |x_1| \leq \epsilon\}$$

SLICES OF THE UNIT BALL

Answer: All but at most a $3e^{-\epsilon^2 d/16}$ fraction of the volume lies within ϵ of the equator.

SLICES OF THE UNIT BALL

Answer: All but at most a $3e^{-\epsilon^2 d/16}$ fraction of the volume lies within ϵ of the equator.



By symmetry, for every unit vector $\mathbf{t}$, the equatorial band is
$$S_{\mathbf{t}} = \{\mathbf{x} \in \mathcal{B}_d : |\mathbf{x}^T \mathbf{t}| \leq \epsilon\}.$$

BIZARRE SHAPE OF THE UNIT BALL

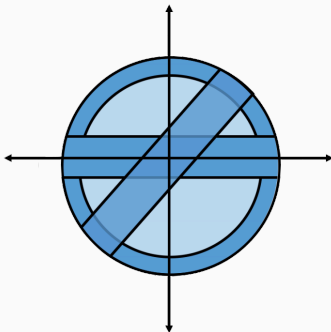
1. At least a $1 - e^{-\epsilon^d}$ fraction of the volume lies within ϵ of the surface.

BIZARRE SHAPE OF THE UNIT BALL

1. At least a $1 - e^{-\epsilon^d}$ fraction of the volume lies within ϵ of the surface.
2. At least a $1 - 3e^{-\epsilon^2 d/16}$ fraction lies within ϵ of any fixed equator.

BIZARRE SHAPE OF THE UNIT BALL

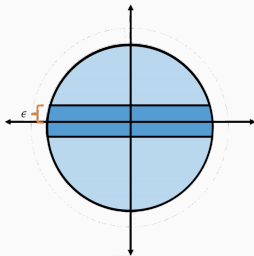
1. At least a $1 - e^{-\epsilon^d}$ fraction of the volume lies within ϵ of the surface.
2. At least a $1 - 3e^{-\epsilon^2 d/16}$ fraction lies within ϵ of any fixed equator.



A high-dimensional ball looks nothing like a two-dimensional ball.

CONCENTRATION AT EQUATOR

Claim: If $\mathbf{x}$ is uniform on $\mathcal{B}_d$, then $|x_1| \leq \epsilon$ with probability at least $1 - 3e^{-\epsilon^2 d/16}$ for $0 < \epsilon \leq 1$.

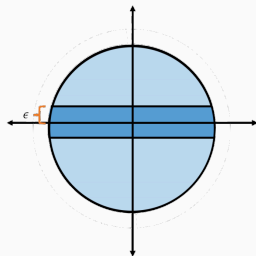


CONCENTRATION AT EQUATOR

Claim: If $\mathbf{x}$ is uniform on $\mathcal{B}_d$, then $|x_1| \leq \epsilon$ with probability at least $1 - 3e^{-\epsilon^2 d/16}$ for $0 < \epsilon \leq 1$.

Proof strategy:

- Replace a random point from the ball by its normalized direction on the unit sphere.

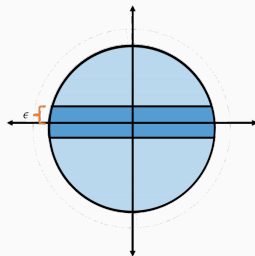


CONCENTRATION AT EQUATOR

Claim: If $\mathbf{x}$ is uniform on $\mathcal{B}_d$, then $|x_1| \leq \epsilon$ with probability at least $1 - 3e^{-\epsilon^2 d/16}$ for $0 < \epsilon \leq 1$.

Proof strategy:

- Replace a random point from the ball by its normalized direction on the unit sphere.
- Let $\mathbf{w} = \mathbf{x}/\|\mathbf{x}\|_2$. Then $\mathbf{w}$ is uniform on the sphere.

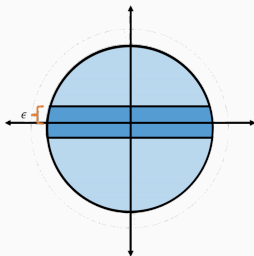


CONCENTRATION AT EQUATOR

Claim: If $\mathbf{x}$ is uniform on $\mathcal{B}_d$, then $|x_1| \leq \epsilon$ with probability at least $1 - 3e^{-\epsilon^2 d/16}$ for $0 < \epsilon \leq 1$.

Proof strategy:

- Replace a random point from the ball by its normalized direction on the unit sphere.
- Let $\mathbf{w} = \mathbf{x}/\|\mathbf{x}\|_2$. Then $\mathbf{w}$ is uniform on the sphere.
- $\Pr[|x_1| \leq \epsilon] \geq \Pr[|w_1| \leq \epsilon]$ (why?)



CONCENTRATION AT EQUATOR

Claim: For $0 < \epsilon \leq 1$, $|w_1| \leq \epsilon$ with probability at least $1 - 3e^{-\epsilon^2 d/16}$. This proves the statement on the previous slide.

CONCENTRATION AT EQUATOR

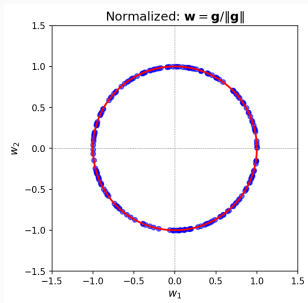
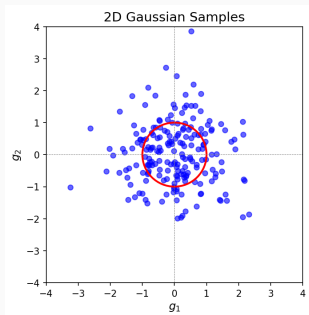
Claim: For $0 < \epsilon \leq 1$, $|w_1| \leq \epsilon$ with probability at least $1 - 3e^{-\epsilon^2 d/16}$. This proves the statement on the previous slide.

How can we generate w , which is a random vector taken from the unit sphere (the surface of the ball)?

CONCENTRATION AT EQUATOR

Claim: For $0 < \epsilon \leq 1$, $|w_1| \leq \epsilon$ with probability at least $1 - 3e^{-\epsilon^2 d/16}$. This proves the statement on the previous slide.

How can we generate w , which is a random vector taken from the unit sphere (the surface of the ball)?



IMPORTANT FACT IN HIGH-DIMENSIONAL GEOMETRY

Rotational invariance of the Gaussian distribution: Let $\mathbf{g}$ be a random Gaussian vector with independent $\mathcal{N}(0, 1)$ entries. Then $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$ is uniform on the unit sphere.

Why?

IMPORTANT FACT IN HIGH-DIMENSIONAL GEOMETRY

Rotational invariance of the Gaussian distribution: Let $\mathbf{g}$ be a random Gaussian vector with independent $\mathcal{N}(0, 1)$ entries. Then $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$ is uniform on the unit sphere.

Why? The density of a high-dimensional Gaussian depends only on $\|\mathbf{g}\|_2$, not on its direction.

IMPORTANT FACT IN HIGH-DIMENSIONAL GEOMETRY

Rotational invariance of the Gaussian distribution: Let $\mathbf{g}$ be a random Gaussian vector with independent $\mathcal{N}(0, 1)$ entries. Then $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$ is uniform on the unit sphere.

Why? The density of a high-dimensional Gaussian depends only on $\|\mathbf{g}\|_2$, not on its direction.

Its probability density function is

$$p(\mathbf{g}) = \prod_{i=1}^d \frac{1}{\sqrt{2\pi}} e^{-g_i^2/2}$$

IMPORTANT FACT IN HIGH-DIMENSIONAL GEOMETRY

Rotational invariance of the Gaussian distribution: Let $\mathbf{g}$ be a random Gaussian vector with independent $\mathcal{N}(0, 1)$ entries. Then $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$ is uniform on the unit sphere.

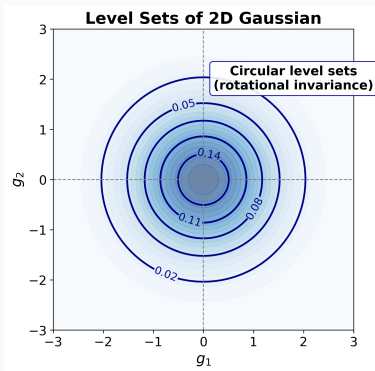
Why? The density of a high-dimensional Gaussian depends only on $\|\mathbf{g}\|_2$, not on its direction.

Its probability density function is

$$\begin{aligned} p(\mathbf{g}) &= \prod_{i=1}^d \frac{1}{\sqrt{2\pi}} e^{-g_i^2/2} \\ &= (2\pi)^{-d/2} e^{-\|\mathbf{g}\|_2^2/2}. \end{aligned}$$

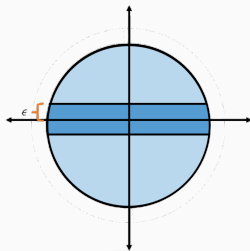
IMPORTANT FACT IN HIGH-DIMENSIONAL GEOMETRY

Rotational invariance of the Gaussian distribution: Let $\mathbf{g}$ have independent $\mathcal{N}(0, 1)$ entries. Then $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$ is uniform on the unit sphere.



PROOF STRATEGY

Draw $\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and set $\mathbf{w} = \mathbf{g} / \|\mathbf{g}\|_2$. We will show that $|w_1| \leq \epsilon$ with high probability.

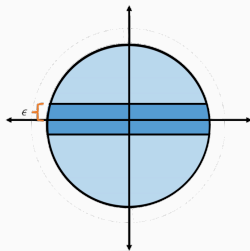


PROOF STRATEGY

Draw $\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and set $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$. We will show that $|w_1| \leq \epsilon$ with high probability.

Two steps:

1. Show that $\|\mathbf{g}\|_2 \geq \sqrt{d/2}$ except with probability $e^{-\Omega(d)}$.

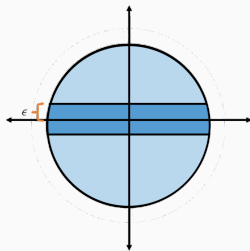


PROOF STRATEGY

Draw $\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and set $\mathbf{w} = \mathbf{g} / \|\mathbf{g}\|_2$. We will show that $|w_1| \leq \epsilon$ with high probability.

Two steps:

1. Show that $\|\mathbf{g}\|_2 \geq \sqrt{d/2}$ except with probability $e^{-\Omega(d)}$.
2. On this event, $|w_1| \leq \sqrt{2/d} |g_1|$. Apply a one-dimensional Gaussian tail bound.



CONCENTRATION AT EQUATOR

Let $\mathbf{g}$ be a random Gaussian vector and $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$.

$$\mathbb{E}[\|\mathbf{g}\|_2^2] = \sum_{i=1}^d \mathbb{E}[g_i^2] = \sum_{i=1}^d \text{Var}[g_i] = d.$$

CONCENTRATION AT EQUATOR

Let $\mathbf{g}$ be a random Gaussian vector and $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$.

$$\mathbb{E}[\|\mathbf{g}\|_2^2] = \sum_{i=1}^d \mathbb{E}[g_i^2] = \sum_{i=1}^d \text{Var}[g_i] = d.$$

Homework exercise: Prove the standard chi-squared lower-tail bound

CONCENTRATION AT EQUATOR

Let $\mathbf{g}$ be a random Gaussian vector and $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$.

$$\mathbb{E}[\|\mathbf{g}\|_2^2] = \sum_{i=1}^d \mathbb{E}[g_i^2] = \sum_{i=1}^d \text{Var}[g_i] = d.$$

Homework exercise: Prove the standard chi-squared lower-tail bound

$$\Pr \left[\|\mathbf{g}\|_2^2 \leq \frac{d}{2} \right] \leq e^{-d/16}.$$

CONCENTRATION AT EQUATOR

Let $\mathbf{g}$ be a random Gaussian vector and $\mathbf{w} = \mathbf{g}/\|\mathbf{g}\|_2$.

$$\mathbb{E}[\|\mathbf{g}\|_2^2] = \sum_{i=1}^d \mathbb{E}[g_i^2] = \sum_{i=1}^d \text{Var}[g_i] = d.$$

Homework exercise: Prove the standard chi-squared lower-tail bound

$$\Pr \left[\|\mathbf{g}\|_2^2 \leq \frac{d}{2} \right] \leq e^{-d/16}.$$

Why should this be plausible?

CONCENTRATION AT EQUATOR

For $1 - 2^{-\Theta(d)}$ fraction of vectors $\mathbf{g}$, $\|\mathbf{g}\|_2 \geq \sqrt{d/2}$. Condition on the event that we get a random vector in this set.

CONCENTRATION AT EQUATOR

For $1 - 2^{-\Theta(d)}$ fraction of vectors $\mathbf{g}$, $\|\mathbf{g}\|_2 \geq \sqrt{d/2}$. Condition on the event that we get a random vector in this set.

Recall that $\mathbf{w} = \frac{\mathbf{g}}{\|\mathbf{g}\|_2}$. Given this event:

$$\Pr[|w_1| \leq \epsilon] = \Pr[|g_1/\|\mathbf{g}\|_2| \leq \epsilon]$$

CONCENTRATION AT EQUATOR

For $1 - 2^{-\Theta(d)}$ fraction of vectors $\mathbf{g}$, $\|\mathbf{g}\|_2 \geq \sqrt{d/2}$. Condition on the event that we get a random vector in this set.

Recall that $\mathbf{w} = \frac{\mathbf{g}}{\|\mathbf{g}\|_2}$. Given this event:

$$\begin{aligned}\Pr[|w_1| \leq \epsilon] &= \Pr[|g_1|/\|\mathbf{g}\|_2 \leq \epsilon] \\ &\geq \Pr\left[|g_1|/\sqrt{d/2} \leq \epsilon\right]\end{aligned}$$

CONCENTRATION AT EQUATOR

For $1 - 2^{-\Theta(d)}$ fraction of vectors $\mathbf{g}$, $\|\mathbf{g}\|_2 \geq \sqrt{d/2}$. Condition on the event that we get a random vector in this set.

Recall that $\mathbf{w} = \frac{\mathbf{g}}{\|\mathbf{g}\|_2}$. Given this event:

$$\begin{aligned}\Pr[|w_1| \leq \epsilon] &= \Pr[|g_1|/\|\mathbf{g}\|_2 \leq \epsilon] \\ &\geq \Pr\left[|g_1|/\sqrt{d/2} \leq \epsilon\right] \\ &= \Pr\left[|g_1| \leq \epsilon \cdot \sqrt{d/2}\right]\end{aligned}$$

CONCENTRATION AT EQUATOR

For $1 - 2^{-\Theta(d)}$ fraction of vectors $\mathbf{g}$, $\|\mathbf{g}\|_2 \geq \sqrt{d/2}$. Condition on the event that we get a random vector in this set.

Recall that $\mathbf{w} = \frac{\mathbf{g}}{\|\mathbf{g}\|_2}$. Given this event:

$$\begin{aligned}\Pr[|w_1| \leq \epsilon] &= \Pr[|g_1|/\|\mathbf{g}\|_2 \leq \epsilon] \\ &\geq \Pr\left[|g_1|/\sqrt{d/2} \leq \epsilon\right] \\ &= \Pr\left[|g_1| \leq \epsilon \cdot \sqrt{d/2}\right] \\ &\geq 1 - 2^{-\Theta\left((\epsilon \cdot \sqrt{d/2})^2\right)} \quad (\text{Integrate the PDF})\end{aligned}$$

CONCENTRATION AT EQUATOR

For $1 - 2^{-\Theta(d)}$ fraction of vectors $\mathbf{g}$, $\|\mathbf{g}\|_2 \geq \sqrt{d/2}$. Condition on the event that we get a random vector in this set.

Recall that $\mathbf{w} = \frac{\mathbf{g}}{\|\mathbf{g}\|_2}$. Given this event:

$$\begin{aligned}\Pr[|w_1| \leq \epsilon] &= \Pr[|g_1|/\|\mathbf{g}\|_2 \leq \epsilon] \\ &\geq \Pr\left[|g_1|/\sqrt{d/2} \leq \epsilon\right] \\ &= \Pr\left[|g_1| \leq \epsilon \cdot \sqrt{d/2}\right] \\ &\geq 1 - 2^{-\Theta\left((\epsilon \cdot \sqrt{d/2})^2\right)} \quad (\text{Integrate the PDF})\end{aligned}$$

CONCENTRATION AT EQUATOR

For $1 - 2^{-\Theta(d)}$ fraction of vectors $\mathbf{g}$, $\|\mathbf{g}\|_2 \geq \sqrt{d/2}$. Condition on the event that we get a random vector in this set.

Recall that $\mathbf{w} = \frac{\mathbf{g}}{\|\mathbf{g}\|_2}$. Given this event:

$$\begin{aligned}\Pr[|w_1| \leq \epsilon] &= \Pr[|g_1|/\|\mathbf{g}\|_2 \leq \epsilon] \\ &\geq \Pr\left[|g_1|/\sqrt{d/2} \leq \epsilon\right] \\ &= \Pr\left[|g_1| \leq \epsilon \cdot \sqrt{d/2}\right] \\ &\geq 1 - 2^{-\Theta\left((\epsilon \cdot \sqrt{d/2})^2\right)} \quad (\text{Integrate the PDF})\end{aligned}$$

By union bound, overall we have:

$$\Pr[|w_1| \leq \epsilon] \geq 1 - 2^{-\Theta(\epsilon^2 d)} - 2^{-\theta(d)}$$

BIZARRE SHAPE OF THE UNIT BALL

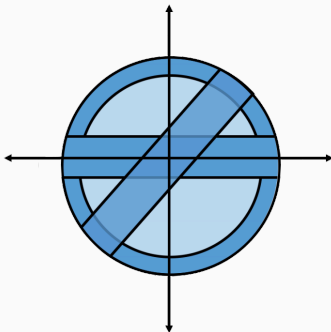
1. At least a $1 - e^{-\epsilon^d}$ fraction of the volume lies within ϵ of the surface.

BIZARRE SHAPE OF THE UNIT BALL

1. At least a $1 - e^{-\epsilon^d}$ fraction of the volume lies within ϵ of the surface.
2. At least a $1 - 3e^{-\epsilon^2 d/16}$ fraction lies within ϵ of any fixed equator.

BIZARRE SHAPE OF THE UNIT BALL

1. At least a $1 - e^{-\epsilon^d}$ fraction of the volume lies within ϵ of the surface.
2. At least a $1 - 3e^{-\epsilon^2 d/16}$ fraction lies within ϵ of any fixed equator.

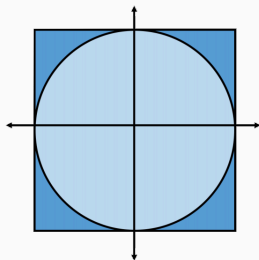


A high-dimensional ball looks nothing like a two-dimensional ball.

HIGH-DIMENSIONAL CUBE

Let $\mathcal{C}_d$ be the d -dimensional cube:

$$\mathcal{C}_d = \{\mathbf{x} \in \mathbb{R}^d : |x_i| \leq 1 \text{ for every } i\}.$$

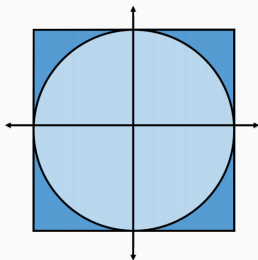


HIGH-DIMENSIONAL CUBE

Let $\mathcal{C}_d$ be the d -dimensional cube:

$$\mathcal{C}_d = \{\mathbf{x} \in \mathbb{R}^d : |x_i| \leq 1 \text{ for every } i\}.$$

In two dimensions, the cube looks broadly similar to the ball.

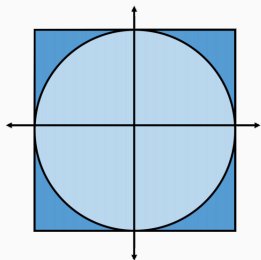


HIGH-DIMENSIONAL CUBE

Let $\mathcal{C}_d$ be the d -dimensional cube:

$$\mathcal{C}_d = \{\mathbf{x} \in \mathbb{R}^d : |x_i| \leq 1 \text{ for every } i\}.$$

In two dimensions, the cube looks broadly similar to the ball.



But their volumes differ dramatically:

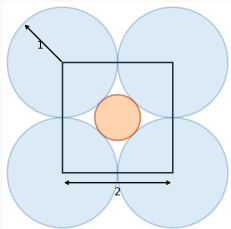
$$\text{vol}(\mathcal{C}_d) = 2^d,$$

$$\text{vol}(\mathcal{B}_d) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)},$$

$$\frac{\text{vol}(\mathcal{C}_d)}{\text{vol}(\mathcal{B}_d)} = d^{\Theta(d)}.$$

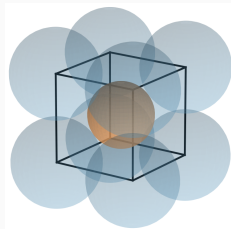
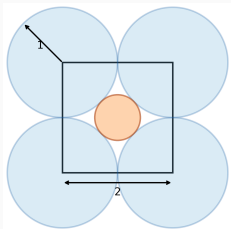
ON BREAKING INTUITION

Consider the following setup:



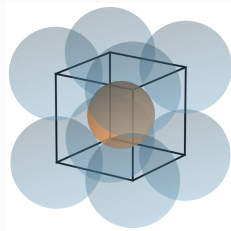
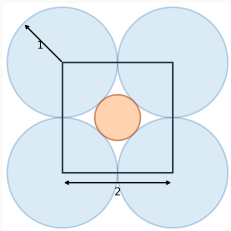
ON BREAKING INTUITION

Consider the following setup:



ON BREAKING INTUITION

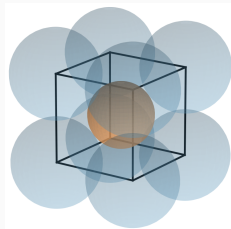
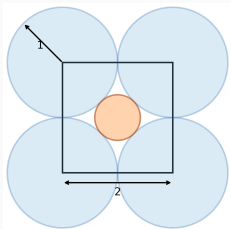
Consider the following setup:



Center 2^d unit balls at the vertices of a side-length-2 cube. The origin-centered ball tangent to them has radius $\sqrt{d} - 1$.

ON BREAKING INTUITION

Consider the following setup:



Center 2^d unit balls at the vertices of a side-length-2 cube. The origin-centered ball tangent to them has radius $\sqrt{d} - 1$.

For $d \geq 5$, this radius exceeds 1, so the central ball protrudes beyond the cube.